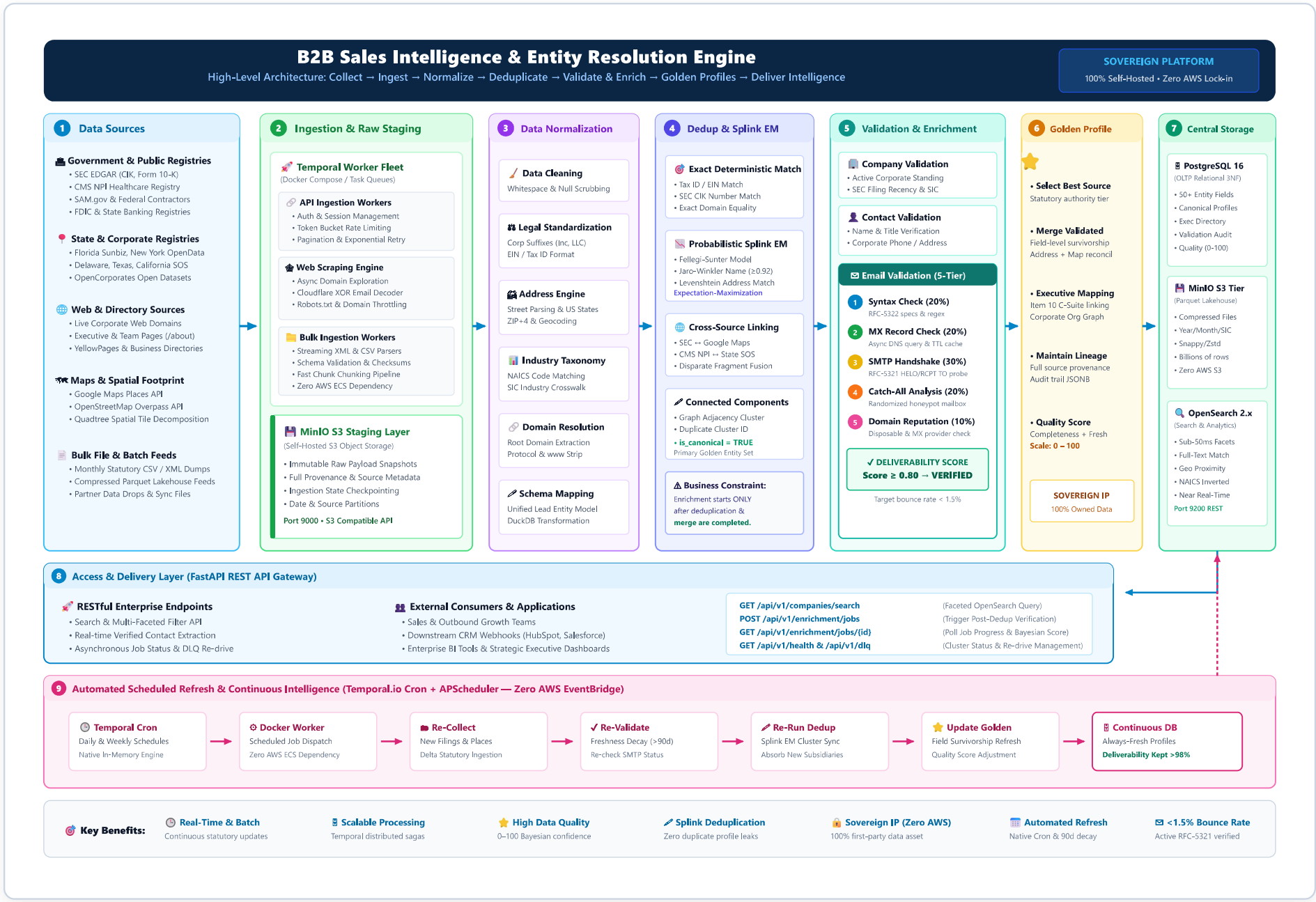

B2B Sales Intelligence & Entity Resolution Engine
High-Level Architecture: Collect → Ingest → Normalize → Deduplicate → Validate & Enrich → Golden Profiles → Deliver Intelligence
SOVEREIGN PLATFORM
100% Self-Hosted • Zero AWS Lock-in
1 Data Sources
Government & Public Registries
• SEC EDGAR (CIK, Form 10-K)
• CMS NPI Healthcare Registry
• SAM.gov & Federal Contractors
• FDIC & State Banking Registries
State & Corporate Registries
• Florida Sunbiz, New York OpenData
• Delaware, Texas, California SOS
• OpenCorporates Open Datasets
Web & Directory Sources
• Live Corporate Web Domains
• Executive & Team Pages (/about)
• YellowPages & Business Directories
Maps & Spatial Footprint
• Google Maps Places API
• OpenStreetMap Overpass API
• Quadtree Spatial Tile Decomposition
Bulk File & Batch Feeds
• Monthly Statutory CSV / XML Dumps
• Compressed Parquet Lakehouse Feeds
• Partner Data Drops & Sync Files
2 Ingestion & Raw Staging
Temporal Worker Fleet
(Docker Compose / Task Queues)
API Ingestion Workers
• Auth & Session Management
• Token Bucket Rate Limiting
• Pagination & Exponential Retry
Web Scraping Engine
• Async Domain Exploration
• Cloudflare XOR Email Decoder
• Robots.txt & Domain Throttling
Bulk Ingestion Workers
• Streaming XML & CSV Parsers
• Schema Validation & Checksums
• Fast Chunk Chunking Pipeline
• Zero AWS ECS Dependency
MinIO S3 Staging Layer
(Self-Hosted S3 Object Storage)
• Immutable Raw Payload Snapshots
• Full Provenance & Source Metadata
• Ingestion State Checkpointing
• Date & Source Partitions
Port 9000 • S3 Compatible API
3 Data Normalization
Data Cleaning
Whitespace & Null Scrubbing
Legal Standardization
Corp Suffixes (Inc, LLC)
EIN / Tax ID Format
Address Engine
Street Parsing & US States
ZIP+4 & Geocoding
Industry Taxonomy
NAICS Code Matching
SIC Industry Crosswalk
Domain Resolution
Root Domain Extraction
Protocol & www Strip
Schema Mapping
Unified Lead Entity Model
DuckDB Transformation
4 Dedup & Splink EM
Exact Deterministic Match
• Tax ID / EIN Match
• SEC CIK Number Match
• Exact Domain Equality
Probabilistic Splink EM
• Fellegi-Sunter Model
• Jaro-Winkler Name (≥0.92)
• Levenshtein Address Match
Expectation-Maximization
Cross-Source Linking
• SEC ↔ Google Maps
• CMS NPI ↔ State SOS
• Disparate Fragment Fusion
Connected Components
• Graph Adjacency Cluster
• Duplicate Cluster ID
• is_canonical = TRUE
Primary Golden Entity Set
⚠ Business Constraint:
Enrichment starts ONLY
after deduplication &
merge are completed.
5 Validation & Enrichment
Company Validation
• Active Corporate Standing
• SEC Filing Recency & SIC
Contact Validation
• Name & Title Verification
• Corporate Phone / Address
Email Validation (5-Tier)
1 Syntax Check (20%)
RFC-5322 specs & regex
2 MX Record Check (20%)
Async DNS query & TTL cache
3 SMTP Handshake (30%)
RFC-5321 HELO/RCPT TO probe
4 Catch-All Analysis (20%)
Randomized honeypot mailbox
5 Domain Reputation (10%)
Disposable & MX provider check
✓ DELIVERABILITY SCORE
Score ≥ 0.80 → VERIFIED
Target bounce rate < 1.5%
6 Golden Profile
• Select Best Source
Statutory authority tier
• Merge Validated
Field-level survivorship
Address + Map reconcil
• Executive Mapping
Item 10 C-Suite linking
Corporate Org Graph
• Maintain Lineage
Full source provenance
Audit trail JSONB
• Quality Score
Completeness + Fresh
Scale: 0 – 100
SOVEREIGN IP
100% Owned Data
7 Central Storage
PostgreSQL 16
(OLTP Relational 3NF)
• 50+ Entity Fields
• Canonical Profiles
• Exec Directory
• Validation Audit
• Quality (0-100)
MinIO S3 Tier
(Parquet Lakehouse)
• Compressed Files
• Year/Month/SIC
• Snappy/Zstd
• Billions of rows
• Zero AWS S3
OpenSearch 2.x
(Search & Analytics)
• Sub-50ms Facets
• Full-Text Match
• Geo Proximity
• NAICS Inverted
• Near Real-Time
Port 9200 REST
8 Access & Delivery Layer (FastAPI REST API Gateway)
RESTful Enterprise Endpoints
• Search & Multi-Faceted Filter API
• Real-time Verified Contact Extraction
• Asynchronous Job Status & DLQ Re-drive
External Consumers & Applications
• Sales & Outbound Growth Teams
• Downstream CRM Webhooks (HubSpot, Salesforce)
• Enterprise BI Tools & Strategic Executive Dashboards
GET /api/v1/companies/search (Faceted OpenSearch Query)
POST /api/v1/enrichment/jobs (Trigger Post-Dedup Verification)
GET /api/v1/enrichment/jobs/{id} (Poll Job Progress & Bayesian Score)
GET /api/v1/health & /api/v1/dlq (Cluster Status & Re-drive Management)
9 Automated Scheduled Refresh & Continuous Intelligence (Temporal.io Cron + APScheduler — Zero AWS EventBridge)
Temporal Cron
Daily & Weekly Schedules
Native In-Memory Engine
Docker Worker
Scheduled Job Dispatch
Zero AWS ECS Dependency
Re-Collect
New Filings & Places
Delta Statutory Ingestion
Re-Validate
Freshness Decay (>90d)
Re-check SMTP Status
Re-Run Dedup
Splink EM Cluster Sync
Absorb New Subsidiaries
Update Golden
Field Survivorship Refresh
Quality Score Adjustment
Continuous DB
Always-Fresh Profiles
Deliverability Kept >98%
Key Benefits:
Real-Time & Batch
Continuous statutory updates
Scalable Processing
Temporal distributed sagas
High Data Quality
0–100 Bayesian confidence
Splink Deduplication
Zero duplicate profile leaks
Sovereign IP (Zero AWS)
100% first-party data asset
Automated Refresh
Native Cron & 90d decay
<1.5% Bounce Rate
Active RFC-5321 verified